USD/CHF Covered Interest Parity Deviations

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Abstract

This note describes monthly estimates of covered interest parity (CIP) deviations for the USD/CHF pair. The measures combine USD/CHF spot and forward exchange rates with short-term USD and CHF interest-rate benchmarks. The resulting series summarize the wedge between the observed interest-rate differential and the differential implied by foreign-exchange hedging markets.

1 Motivation

Covered interest parity is a central no-arbitrage condition in international finance. In frictionless markets, borrowing in one currency, swapping the proceeds through the forward market, and investing in another currency should deliver the same hedged return as investing domestically. When this condition fails, the resulting cross-currency basis measures the price of synthetic funding across currencies. Persistent deviations are therefore informative about limits to arbitrage, balance-sheet constraints, funding pressures, regulatory costs, and the relative scarcity of safe assets (Akram, Rime, and Sarno, 2008; Borio et al., 2016; Du, Tepper, and Verdelhan, 2018).

The USD/CHF pair is a useful object for this exercise. The U.S. dollar is the dominant international funding and invoicing currency, while the Swiss franc is a safe-haven currency associated with deep financial markets and long periods of very low or negative interest rates. Deviations from CIP in this pair therefore offer a compact measure of stress and segmentation between dollar funding markets, Swiss-franc money markets, and foreign-exchange hedging markets. The comparison with the CHF government-bond CIP benchmark of Du, Keerati, and Schreger (2020) is useful because it relates the short-rate estimates to a closely related government-yield benchmark for the same currency pair and tenor.

2 Definition

Let S_t denote the spot exchange rate quoted in CHF per USD, and let $F_t^{(T)}$ denote the corresponding forward exchange rate for tenor T. With this quote convention, the forward-implied USD minus CHF interest-rate differential is

$$\widehat{r}_{USD,t}^{(T)} - \widehat{r}_{CHF,t}^{(T)} = -\frac{1}{T} \log\left(\frac{F_t^{(T)}}{S_t}\right). \tag{1}$$

The CIP deviation is defined as the difference between the observed interest-rate differential and the forward-implied differential:

$$b_t^{(T)} = \left(r_{USD,t}^{(T)} - r_{CHF,t}^{(T)}\right) - \left[-\frac{1}{T} \log\left(\frac{F_t^{(T)}}{S_t}\right)\right]. \tag{2}$$

The series report $10000 \times b_t^{(T)}$, so values are annualized basis points. Positive values indicate that the observed USD-minus-CHF interest-rate differential is above the differential implied by the forward market; negative values indicate the opposite.

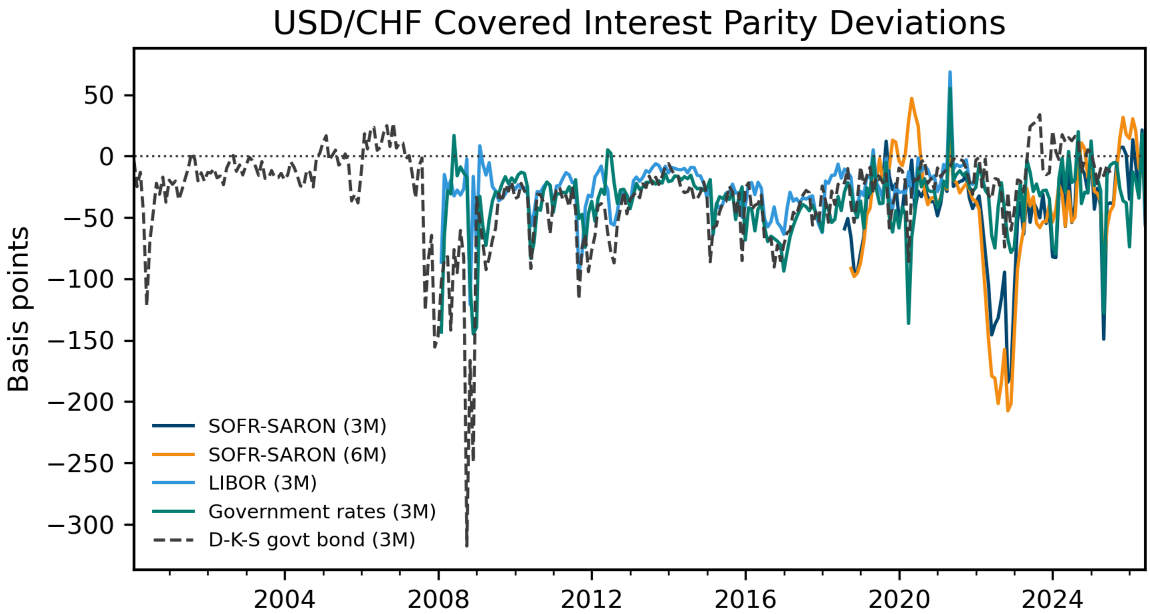
3 Data and Measures

The exchange-rate data are from the Swiss National Bank and include USD/CHF spot, three-month forward, and six-month forward rates. The modern risk-free-rate measures use SOFR from FRED and SARON compound rates from the Swiss National Bank. The legacy measure uses USD and CHF LIBOR series. The government-rate measure combines the 3-month U.S. Treasury constant-maturity yield from FRED with the Swiss Confederation money-market series from the Swiss National Bank.

The public data file reports four estimated deviation series: the three-month SOFR-SARON basis, the six-month SOFR-SARON basis, the three-month LIBOR basis, and the three-month government-rate basis. The figure also plots the sign-adjusted CHF government-bond CIP benchmark of Du, Keerati, and Schreger (2020). In that benchmark, the published convention is reversed relative to the USD-minus-CHF definition used here, so the series is multiplied by -1 before being plotted.

4 Chart

Figure 1 plots the monthly basis measures in basis points.



Source: Author's calculations using Swiss National Bank and FRED data; benchmark from Du, Keerati & Schreger (2020).

Figure 1: Monthly USD/CHF covered interest parity deviations

5 Replication

The accompanying code implements the definitions above using public data from the Swiss National Bank, FRED, and the Du-Keerati-Schreger benchmark dataset. The repository is designed to update the monthly estimates and figure automatically.

References

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